



Ahamed Sazjaath MOHAMED

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WORK EXPERIENCE

Centre Inria Université Grenoble Alpes – Grenoble, France

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Engineer in robotics

[1 Oct 2024 – Current]

- Finding solutions to necessities for robotic applications based on **ROS2**.
- Implementing **Visual-SLAM** in humanoid robot for precise **localization**.
- Constructing **Semantic Maps** to enable high-level, natural language-based vocal commanding.
- Programming a new **Nav2** controller based on **Deep Reinforcement Learning** to optimize computational load in the robot.
- Training and programming **deep neural networks** for robotic applications.

Alten SO, Alten Labs – Toulouse, France

Website: Alten.fr

Intern in robotics

[5 Feb 2024 – 2 Aug 2024]

- Final year project on **robot fleet management** for Industry 5.0
- Create a Simulation using **ROS2** and Gazebo simulator.
- The simulation aims to simulate a warehouse equipped with multiple robots to optimize fabrication line, analyze and compare **task allocation algorithms** to optimally manage robotic fleet.
- Research on **optimal path planning and control** algorithms for **multi robot navigation**.

Deltacad – Lacroix St-Ouen, France

Website: deltacad.fr

Engineering assistant - Intern

[4 Sep 2023 – 16 Feb 2024]

- Assistant Engineer Internship.
- **Developed** a full-stack defect detection prototype application, featuring a **C# GUI** and backend detection algorithms implemented in **C++ and Python** (using OpenCV).
- **Engineered** an object segmentation and **normalization/rescaling** algorithm to **standardize** input images, creating a robust dataset that significantly **enhanced** the performance of subsequent **Deep Neural Network (DNN)** detection models.

EDUCATION AND TRAINING

Computer Science Engineering degree

Université de Technologie Compiègne [2 Sep 2019 – 2 Aug 2025]

City: Compiègne | Country: France | Website: <https://www.utc.fr/formations/diplome-dingenieur/genie-informatique-gi/filiere-informatique-embarquee-et-systemes-autonomes-ines/> | Field(s) of study: Engineering,

manufacturing and construction: • Electronics and automation | **Level in EQF:** EQF level 7 |
Type of credits: ECTS

- Computer science and robotic systems.
- Intelligent vehicles.
- Multi-agent systems in robotics.
- Perception and sensor fusion for localization.

Baccalaureat Générale

Lycée Marie Curie [4 Sep 2017 – 5 Jul 2019]

City: Nogent-sur-Oise | **Country:** France | **Level in EQF:** EQF level 4

PROJECTS

[4 Sep 2023 – 19 Jan 2024]

Sensor fusion for precise localisation in intelligent vehicles

- Academic group project.
- Estimate precise location of a vehicle to avoid using Real Time Kinematics (RTK).
- Kalman filter to fusion the sensor data from multiple sensors like GPS, IMU, LiDAR, and images from cameras.

[4 Sep 2023 – 19 Jan 2024]

Multi-agent systems

- Academic group project.
- Project done with robotarium simulator framework in Matlab.
- Multi-robot localization and Control in a specific leader-follower formation.
- Litterature review for adapted controller for a smooth navigation with robots.

ACADEMIC MENTORSHIP

[5 May 2025 – 18 Jul 2025]

Technical Mentorship: Robotic Pose Estimation

- Supervised a Master 1 student in the research and deployment of efficient pose estimation models tailored for [Company/Lab Name] robots.
- Guided the student from initial literature review to final implementation, offering critical analysis and troubleshooting for complex spatial computing challenges.
- Result: Successfully optimized the pose estimation pipeline, reducing the Deep Neural Network (DNN) GPU footprint by 50% while maintaining real-time performance on robotic hardware.

LANGUAGE SKILLS

Mother tongue(s): Tamil

Other language(s):

English

LISTENING C1 **READING** C2 **WRITING** C1
SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

French

LISTENING C1 **READING** C1 **WRITING** B2
SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Spanish

LISTENING A2 **READING** A2 **WRITING** A2
SPOKEN PRODUCTION A2 **SPOKEN INTERACTION** A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Knowledge of Python, MATLAB, and C/C++ coding languages / Databases (MongoDB MySql PostgresSql) / C# / CAD-Modelling / Experience with Robot Operating System (ROS) and Linux (ROS1 and ROS2) / Linux / Nav2 / linux shell scripting / Embedded Systems: Arduino, NodeMCU, and other microcontrollers / Deployment on RaspberryPi / Basic knowledge of Robotics / Version Control (Git) / Docker Container Architecture / Sensors' data processing / C/C++ / Microsoft Office, LibreOffice, LaTeX / Docker / SVN / Typst

PUBLICATIONS

[2026]
OpenSocInt: A Multi-modal Training Environment for Human-Aware Social Navigation In this paper, we introduce OpenSocInt, an open-source software package providing a simulator for multi-modal social interactions and a modular architecture to train social agents. We described the software package and showcased its interest via an experimental protocol based on the task of social navigation. Our framework allows for exploring the use of different perceptual features, their encoding and fusion, as well as the use of different agents

Authors: Victor Sanchez, Chris Reinke, Ahamed Mohamed, Xavier Alameda-Pineda | **Publisher:** ArXiv

Victor Sanchez et al. , 2026, OpenSocInt: A Multi-modal Training Environment for Human-Aware Social Navigation, ArXiv

RECOMMENDATIONS

Name: Xavier Alameda Pineda | Research director and leader of team RobotLearn at Inria Grenoble



Email: xavier.alameda-pineda@inria.fr

Name: Lounis Adouane | Professor at Université de Technologie Compiègne



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